

Predicting Drug Blood-Brain Barrier Penetration with Adverse Event Report
Embeddings
--Manuscript Draft--

Manuscript Number:	JB1-21-293
Article Type:	Research paper
Keywords:	Blood Brain Barrier; FDA Adverse Event Reports; Drug Embeddings; Natural Language Processing
Abstract:	Adverse event reports (AER) are widely used for post-market drug safety surveillance, and in recent work have also been used as a source of information for drug repurposing, with the assumption that drugs with similar side-effects may have similar therapeutic effects also. In this study, we used distributed representations of drugs derived from the Food and Drug Administration (FDA) AER system using aer2vec, a method of representing AER, with drug embeddings emerging from a neural network trained to predict the probability of adverse drug effects given observed drugs. We combined these representations with molecular features to predict permeability of the blood-brain barrier to drugs, a prerequisite to their application to treat conditions of the central nervous system. Using logistic regression classifiers, the addition of distributed representations improved performance over prior methods using drug-drug similarity estimates derived from discrete representations of AER system data. This was true both in cross-validation experiments with two reference sets, and in a cross-training configuration involving training on one set and testing on the other. All approaches evaluated using embedding-based features outperformed approaches using discrete statistically-derived drug similarities as features, with the best individual model attaining AUCs of 0.81 and 0.87 on the two reference sets used for evaluation, corresponding to improvements in absolute AUC of ~5% and ~9% over the best-performing models using statistically-derived similarity, and 9% and 13% over performance with molecular features only. Performance advantages were retained in cross-training experiments, and when reducing embedding dimensions from 500 to 6, indicating that they are neither attributable to overfitting, nor to a difference in the number of trainable parameters. These results indicate that aer2vec distributed representations carry information that is valuable for predicting therapeutic effects in addition to their established utility for anticipating the harmful effects of drugs.



Mar 30, 2021

Mor Peleg, PhD
Editors-in-Chief
Journal of Biomedical Informatics

Dear Dr. Peleg,

Enclosed, please find an original research article titled, "Predicting Drug Blood-Brain Barrier Penetration with Adverse Event Report Embeddings," written with my coauthors Dr. Justin Mower, Dr. Devika Subramanian, and Dr. Trevor Cohen, for possible publication in the *Journal of Biomedical Informatics (JBI)*.

In this article, we used distributed representations of drugs derived from the Food and Drug Administration (FDA) adverse event (AER) system using aer2vec, a method of representing AER, with drug embeddings emerging from a neural network trained to predict the probability of adverse drug effects given observed drugs. We combined these representations with molecular features to predict the permeability of the blood-brain barrier to drugs, a prerequisite to their application to treat conditions of the central nervous system. Using machine learning classifiers in the context of two evaluation sets, the addition of distributed representations improved performance over prior methods using drug-drug similarity estimates derived from discrete representations of AER system data. These results indicate that aer2vec distributed representations carry information that is valuable for predicting therapeutic effects in addition to their established utility for anticipating the harmful effects of drugs. As such, we anticipate the paper will be of interest to members of the JBI readership with interest in applying semantic representation derived from the FDA AER system as a means to improve the performance of predictive models.

We initially submitted the manuscript to the journal *Bioinformatics*, where some of the work we drew inspiration from was originally published. Unfortunately, our paper was rejected with scant feedback from reviewers to explain the justification for this decision. We suspect that our emphasis on the comparison between representational approaches may be better suited to JBI. Nonetheless, we have included the reviewers' comments, together with our responses to them, in the submission package. This decision notwithstanding, we remain enthusiastic about the quality of this work, and its implications for

the field.

Thank you for taking the time to consider this manuscript. We look forward to hearing from you, and please do not hesitate to contact me if further information is required.

Sincerely,

A handwritten signature in blue ink, appearing to read 'YiFan Wu', with a long horizontal flourish extending to the right.

YiFan Wu

We initially submitted the manuscript to the journal *Bioinformatics*, where some of the work we drew inspiration from was originally published. Please find below critiques raised by reviewers in response to the original submission with our point-by-point responses in red.

Reviewer: 1

Comments to the Author

General comments:

In this manuscript, the authors adopted `aer2vec` and molecular features to learn combined embeddings of drug molecules, and then used learned drug representations to predict permeability of the blood-brain barrier (BBB) to drugs. The `aer2vec` approach learned both drug and adverse drug event (ADE) embeddings by predicting probability of an ADE under a given drug context. On the other hand, XlogP and molecular weight of BBB drugs were selected as molecular properties; and similarities were computed by an expert-curated drug panel and aer2vec embeddings, resulting in six statistical features and distributed features. Model were based on logistic regression (LR) with L2-regularization. Experiments under various dataset settings showed that this workflow achieved better performance than those that only used chemical features (CF) or CF with expert-curated similarities. Furthermore, experiments also demonstrated that the similarities and the principal components based on `aer2vec` embeddings also made higher AUC. This workflow can help researchers check the permeability of the blood-brain barrier to drugs and find potential drugs for the central nervous system.

Major:

The permeability of the blood-brain barrier to drugs can be considered as one type of biological property, like toxicity. The SMILES strings and molecular fingerprints of drug compounds are also easily accessible, and there have been some works based on similarities according to molecular fingerprints for BBB prediction [1]. How about comparing this method with such previous works? Although the experiments in this paper were well-designed, the method may lack novelty, since the embedding algorithm `aer2vec` was proposed by Portanova et al., and the predictive LR model was inherited from McCoy and Perlis.

[1] Wang, Z., Yang, H., Wu, Z., Wang, T., Li, W., Tang, Y., & Liu, G. (2018). In Silico Prediction of Blood-Brain Barrier Permeability of Compounds by Machine Learning and Resampling Methods. *ChemMedChem*, 13(20), 2189–2201. <https://doi.org/10.1002/cmdc.201800533>

Response:

Thank you for raising this issue, and drawing our attention to this paper. While we are aware that the features that have been used for predicting blood-brain barrier permeability extend beyond side-effect profiles, the purpose of our work was not to compare the relative utility of these features. Rather, we wished to make a meaningful comparison of the utility of different representations (discrete statistical vs. distributed neural embeddings) of the same data for this purpose. We chose side effect profiles on account of promising prior work described in our

paper, and endeavored to reproduce this work with the sole change of substituting **one representation of adverse event reports** for the other. While there may be ways to make similar comparisons on the basis of SMILES strings or molecular fingerprints, we view this as outside the scope of the current work. In particular, these data sources do **not** carry **distributional information** – strings or molecular features are either present or absent, whereas co-occurrence events in adverse event data are repeated for frequently occurring events. Thus, the central question of our work – whether or **not distributed representations** of the same distributional information offer advantages over discrete representations cannot be addressed in the context of these other data sources. In the revised manuscript we explicitly acknowledge this as a (deliberate) limitation of our work and discuss evaluating the potential utility of incorporating other data as a direction for future work (Limitation: Page 11, paragraph 2).

Reviewer: 2

Comments to the Author

The authors carried out this research through the hypothesis of "adding **distributed drug vectors** generated from **FAERS** data to the basic chemical properties would improve performance when identifying the permeability of the **BBB** to a drug" and verified that this hypothesis is indeed desirable. But there are still **some flaws** in the article that need to be improved.

1. The description of part of the content of the article is cumbersome and does **not** have good **readability**. It is recommended that the author can illustrate the research process and content in the form of a flowchart to help readers understand. In addition, many acronyms are used in the article. A moderate amount of acronyms is helpful for reading the article, but too many acronyms increase the reader's reading burden.

Response:

Thank you for your response. In the revised manuscript we have followed your suggestion, adding a **flowchart to the Methods** section (page 7) to help readers better understand the overall flow of our project. In addition, we revised the paper to make sure definitions were provided for all acronyms when they are first introduced, and when they are used in tables and figures.

2. The subsection 2.1.3 of the article is titled "Chemical Properties", but the author is introducing the origin of the data in the entire subsection.

Response:

Thanks for pointing that out. We have changed the title of the subsection to "Reference Sets and Data Sources" (page 5, section 2.1.3)

3. Page 4, there is a sentence "The vectors were trained over 25 iterations across the data with five negative samples per positive sample". I cannot understand the meaning of "the data with five negative samples per positive sample".

Response:

We have added the following content to explain this detail of the methods (page 5, paragraph 1):

The **aer2vec** model is trained to maximize the predicted probabilities for observed **drug/ADE co-occurrence** events (where a drug and ADE appear in the same FAERS report) while minimizing this predicted probability for randomly sampled counterexamples (the 'negative samples' in skipgram-with-negative-sampling). For each observed drug/ADE co-occurrence, we randomly draw **five** other ADEs, and minimize the predicted probability of their co-occurring with the drug in question. We used five negative samples per observed co-occurrence event, following Portanova et al. (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7153155/>)

4. A 500-dimensional distributed representation is used as a feature in the research, but the **origin and general** content of the 500-dimensional are not explained.

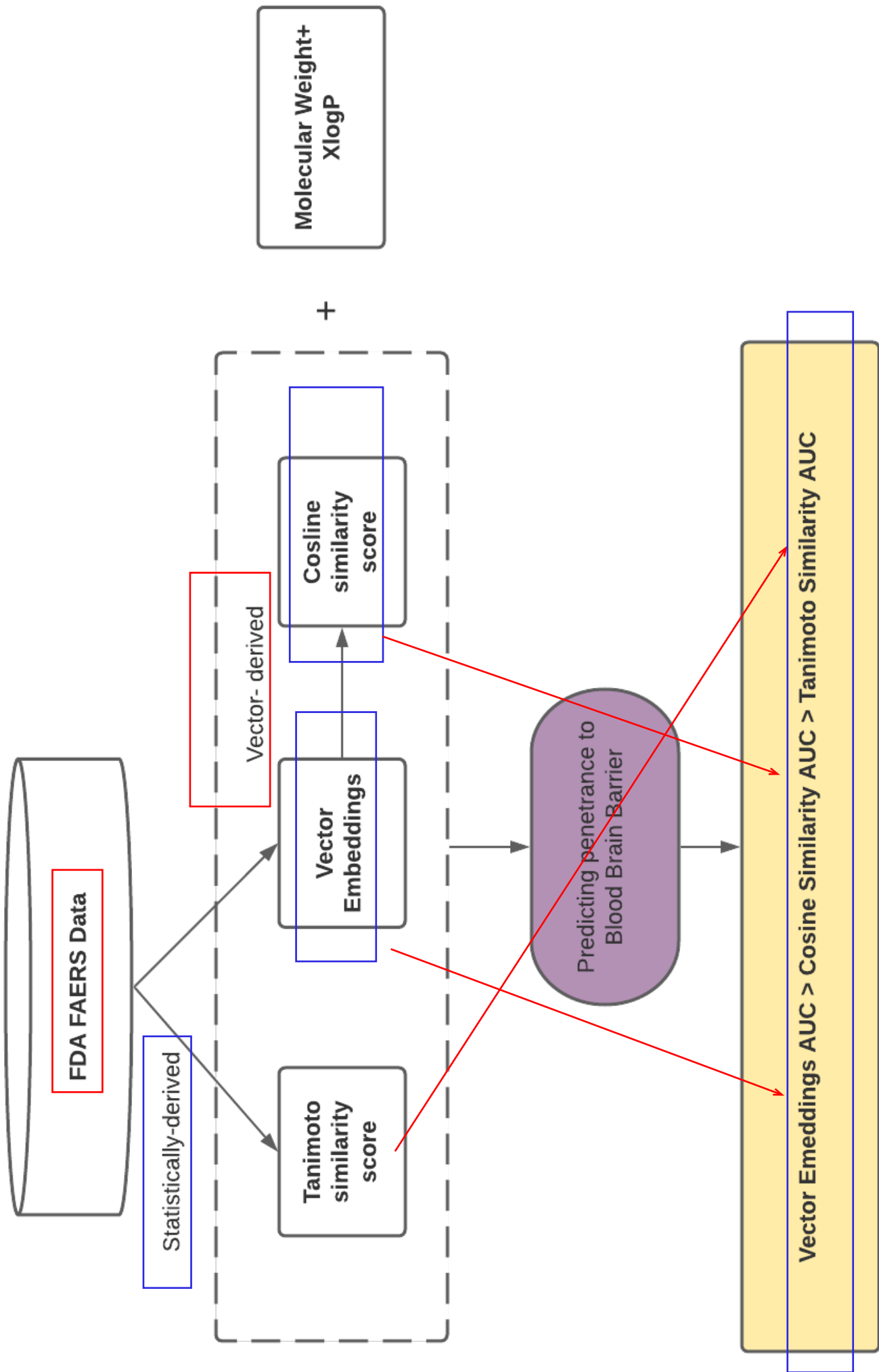
Response:

The 500-dimensional vectors were generated using the aer2vec algorithm of Portanova and colleagues, which derives neural drug and adverse drug event representations from adverse event report data. This algorithm is described in detail in our manuscript in section 2.1.1 and 2.1.2 (page 4). We also provide a link to the **aer2vec** GitHub repository (the original manuscript can be found at <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7153155/>), which contains code and instructions through which interested readers may build their own **aer2vec** models.

5. Most of the content of this research is based on the **existing research methods**, and the follow-up **comparative experiments** are less content, which cannot reflect the **workload** of the authors

Response:

While it is true that we apply the **previously developed aer2vec method**, we would argue that we are applying it in the context of a **novel domain** to address previously **unanswered questions**. Specifically, while aer2vec has been evaluated as a means to identify probable drug/side-effect relationships, it has **not** previously been evaluated as a method for drug **repurposing**. The current work is the first to explore this possibility, in an effort to determine whether or not the established advantages of aer2vec for pharmacovigilance extend to this problem domain also. Furthermore, **aer2vec** results in prior work have **depended upon recapitulating the probabilistic training objective as a model**. This work is the first to **evaluate the utility of the resulting embeddings in the context of a downstream supervised machine learning task**.



Highlights

Predicting Drug Blood-Brain Barrier Penetration with Adverse Event Report Embeddings

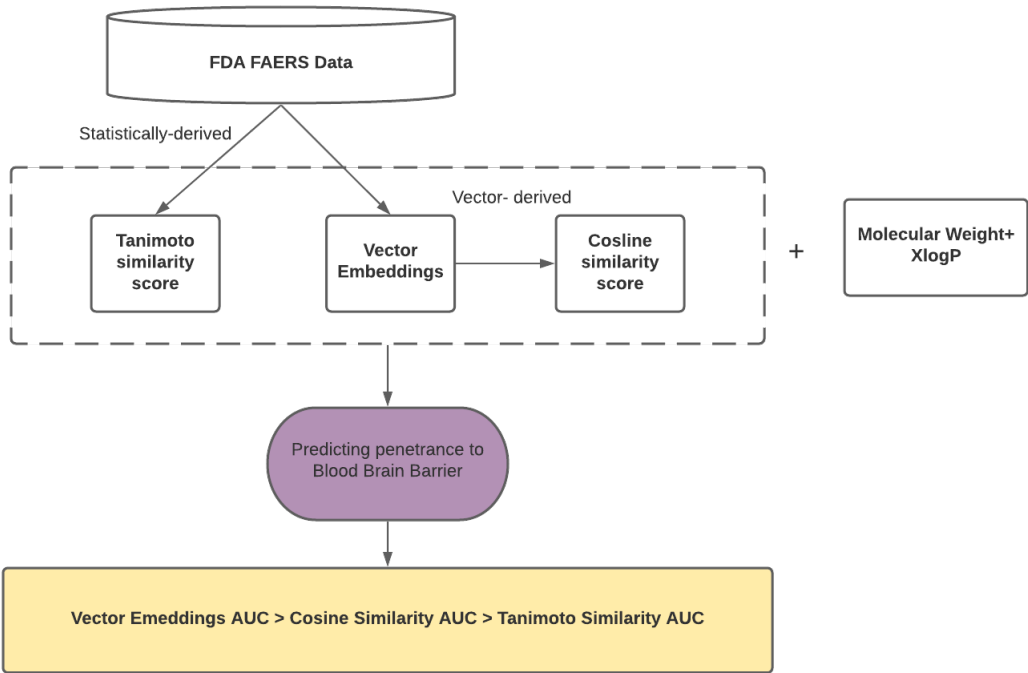
YiFan Wu, Justin Mower, Devika Subramanian, Trevor Cohen

- FDA Adverse Event Report derived embeddings carry information of value for predicting therapeutic effects in addition to their established utility for anticipating harmful side effects.
- The drug embeddings derived from adverse event report outperformed statistically-derived drug-drug similarity scores in prediction of drug penetrance to blood brain barrier.
- This advantage in performance is retained when reducing drug embeddings to a dimensionality consistent with that of models employing statistically-derived similarity, and is therefore not attributable to differences in the number of trainable parameters.

Graphical Abstract

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Highlights

Predicting Drug Blood-Brain Barrier Penetration with Adverse Event Report **Embeddings**

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Predicting Drug Blood-Brain Barrier Penetration with Adverse Event Report Embeddings^{*}

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ARTICLE INFO

Keywords:

Drug Repurposing

Blood Brain Barrier

FDA Adverse Event Reports

Drug Embeddings

Natural Language Processing

ABSTRACT

Adverse event reports (AER) are widely used for post-market drug safety surveillance, and in recent work have also been used as a source of information for drug repurposing, with the assumption that drugs with similar side-effects may have similar therapeutic effects also. In this study, we used distributed representations of drugs derived from the Food and Drug Administration (FDA) AER system using *aer2vec*, a method of representing AER, with drug embeddings emerging from a neural network trained to predict the probability of adverse drug effects given observed drugs. We combined these representations with molecular features to predict permeability of the blood-brain barrier to drugs, a prerequisite to their application to treat conditions of the central nervous system. Using logistic regression classifiers, the addition of distributed representations improved performance over prior methods using drug-drug similarity estimates derived from discrete representations of AER system data. This was true both in cross-validation experiments with two reference sets, and in a cross-training configuration involving training on one set and testing on the other. All approaches evaluated using embedding-based features outperformed approaches using discrete statistically-derived drug similarities as features, with the best individual model attaining AUCs of 0.81 and 0.87 on the two reference sets used for evaluation, corresponding to improvements in absolute AUC of ~5% and ~9% over the best-performing models using statistically-derived similarity, and ~9% and ~13% over performance with molecular features only. Performance advantages were retained in cross-training experiments, and when reducing embedding dimensions from 500 to 6, indicating that they are neither attributable to overfitting, nor to a difference in the number of trainable parameters. These results indicate that *aer2vec* distributed representations carry information that is valuable for predicting therapeutic effects in addition to their established utility for anticipating the harmful effects of drugs.

1. Introduction

Drug development is a long and expensive process. The time from drug development to market release takes on average 14 years, with a median cost of \$19 million among 59 approved drugs from 2015-2016 (Moore, Zhang, Anderson and Alexander, 2018). The repurposing of drugs that have already been approved for some other use is a desirable strategy because their safety and efficacy have been established previously, shortening the development pipeline, and this strategy has already been used to identify new usages of approved drugs (Kumar, Harilal, Gupta, Jose, Thomas Parambi, Uddin, Shah and Mathew, 2019; Ashburn and Thor, 2004). For instance, the repurposing of the anti-diabetic agent *metformin* for cancer and for cardiovascular diseases are successful examples of drug repurposing (Rena and Lang, 2018; Heckman-Stoddard, DeCensi, Sahasrabuddhe and Ford, 2017). Both computational (in-silico) and experimental (in-vitro) methods have been applied to drug repurposing (Pushpakom, Iorio, Eysers, Escott, Hopper, Wells, Doig, Guillems, Latimer, McNamee, Norris, Sanseau, Cavalla and Pirmohamed, 2019). With advances in machine learning methods and processing power, computational approaches have gained in popularity. A variety of techniques have been devised for computational drug repurposing using data sources such as large dimensional -omics data and retrospective clinical trial data amongst others (Pushpakom et al., 2019). An additional data source that has been explored recently consists of the side-effect profiles of drugs, with the assumption that drugs that have similar side effects may have similar therapeutic effects also (Wang, Wan, Wang and Weng, 2016).

^{*} This document is the results of the research project funded by National Library of Medicine Grant R01LM011563 and 5T15LM007442-18.

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One source of side-effect profiles is adverse event report (AER) data. An adverse event of a drug is defined as an injury resulting from a medical intervention that uses it (Edwards and Aronson, 2000). It is estimated that adverse events of drugs account for 5% of complications after discharge, with data drawn from several international studies (Hennessy and Strom, 2015; Campillos, Kuhn, Gavin, Jensen and Bork, 2008). Therefore, monitoring the post-market effects of pharmaceutical products is critical to ensure the safety of patients. To this end, the United States Food and Drug Administration (FDA) has maintained the FDA Adverse Event Reporting System (FAERS) for more than 50 years. FAERS receives over a half-million reports annually (Harpaz, DuMouchel, LePendou, Bauer-Mehren, Ryan and Shah, 2013). Each report includes the drugs a patient was taking when an adverse drug event (ADE) was observed, with reporters subjectively classifying putative causative agents as “primary suspects” (PS) or “secondary suspects” (SS) with presumed bystanders designated as “concomitant” (C). CNS-active agents are well-represented in FAERS. For example, in a 2015 drug-safety surveillance report, the authors noted that the most frequently reported class in FAERS was “nervous system drugs” (58%), and that the second and third most commonly reported system classes were “nervous system” (12%) and “psychiatric” (11%) in pediatric regimens (de Bie, Ferrajolo, Straus, Verhamme, Bonhoeffer, Wong, Sturkenboom, Sturkenboom, Bonhoeffer, Giaquinto, Gazarian, Ceci, Weereasuriya, Hirschfeld, Saint Raymond, Nakamura, Sharland, Offringa, Wong and Hoppu, 2015). AER data are primarily used for post-marketing drug safety surveillance. To this end, researchers have established data mining methodologies, known as Signal Detection Algorithms (SDAs), that leverage AER data for pharmacovigilance signal detection. The FDA utilizes SDAs, such as disproportionality analysis (DPA) methods and multi-variate modeling techniques, to conduct quality control over the spontaneous reports in FAERS and prioritize safety surveillance signals for consideration in the regulation of drugs after their release to market (Bate and Evans, 2009; Harpaz et al., 2013).

The utility of side effect patterns as a means to identify new therapeutic indications has only recently been evaluated. A first step toward this goal involves estimation of the similarity between pairs of drugs, based on their side effect profiles. For example, Tatonetti et al. have constructed a database, OffSides, consisting of adverse effects and drug pairs that have strong signal as estimated using SDAs applied to FDA post market ADE reports (Tatonetti, Ye, Daneshjou and Altman, 2012b). The authors derived a similarity score between two drugs following a method originally described by Campillos et al. in (Campillos et al., 2008). This work demonstrated that two drugs eliciting similar ADEs have a high likelihood of interacting with a common drug target (Campillos et al., 2008). To facilitate the estimation of similarity between drugs, drugs were representing as binary (bit) vector such that each dimension represents the presence or absence of a significant association with a particular side effect. Significance here was determined by calculating the proportional reporting ratio (PRR) between each drug each adverse event in FAERS, and retaining PRRs above a threshold of statistical significance.

The drug similarity score was measured using the Tanimoto coefficient (equation 1), where N_a represents the set (i.e. “one”) bits in the bit vector representing drug a and N_b represents these bits for drug b (Rogers and Tanimoto, 1960). The Tanimoto coefficient is then calculated as the ratio of the set bits in common to the total number of set bits across both drugs, effectively the intersection ($\cap$) over the union ($\cup$) of the set of side effects significantly associated with each drug. Since some drugs had higher mean similarity scores than the others, a Z-transformation of the similarities for each drug was used to normalize the scores (Tatonetti et al., 2012b).

$$TC = \frac{|N_a \cap N_b|}{|N_a| + |N_b| - |N_a \cap N_b|} \quad (1)$$

McCoy and Perlis adopted the similarity scores derived by Tatonetti et al. (which are in turn derived from discrete counts of side effects in FAERS) to predict the permeability of the blood-brain barrier (BBB) to drugs (McCoy and Perlis, 2015). As a part of the central nervous system (CNS), the BBB is a highly selective barrier that only approximately 2% of molecules in the random chemical pool can pass (Gao, Chen, Cai and Xu, 2017; Miao, Xia, Chen, Huang and Liang, 2019). Treatments for psychiatric and other CNS disorders must cross the BBB to exert psychopharmacological effects. However, it has been difficult to find economical high-throughput approaches to estimate brain penetrance (McCoy and Perlis, 2015). In-silico prediction of the permeability of the BBB would provide an appealing solution to the problem of anticipating whether or not a drug might be effective as a CNS agent (Gao et al., 2017). Furthermore, previous research has demonstrated the feasibility of using both machine learning and conventional statistical approaches to predict the permeability of the BBB to particular molecules (Gao et al., 2017; McCoy and Perlis, 2015; Miao et al., 2019; Wang, Yang, Wu, Wang, Li, Tang and Liu, 2018; Doniger, Foxe, Murray, Higgins and Javitt,

2002).

Previous work in this area has found that compounds with a $P(\text{octanol/water})$ coefficient (referred to as $XlogP$, or the *lipophilicity* coefficient) close to 3.4, a molecular weight of less than 400-600 Da, and a sum of nitrogen and oxygen atoms less than or equal to 5 have a higher probability of passing the BBB (Kortagere, Chekmarev, Welsh and Ekins, 2008; Wang et al., 2018). Additionally, sophisticated models that incorporate multiple chemical properties of drugs such as lipophilicity, molecular weight, the topological polar surface area, hydrogen bond donors, receptors and molecular descriptors and fingerprints have been successfully developed (Bergstrom, Charman and Nicolazzo, 2012). However, these models are restricted by the availability of their key parameters, and consequently are difficult for investigators to apply to screen for potentially useful agents at scale.

In 2017, Gao et al. developed a method to predict the permeability of the BBB to both micro- and macro-molecules, outperforming conventional chemical feature based methods. Gao et al. use a support vector machine (SVM) classifier with Monte Carlo cross-validation trained on nine physicochemical and clinical features across two datasets. Evaluation was conducted using a manually curated dataset containing 213 drugs, and the authors report an accuracy of 76% and an AUC of 0.739 (Gao et al., 2017). In two recent studies, Miao et al. and Wang et al. applied deep learning to predict the permeability of the BBB to chemical compounds. Both of these recent studies reported high AUC scores ($AUC > 0.9$) (Miao et al., 2019; Wang et al., 2018). However, as neither the code nor the data from these experiments are publicly available, these results are not readily reproducible.

Of particular relevance to the current research, McCoy and Perlis leveraged side effect profiles to predict BBB permeability (McCoy and Perlis, 2015). In this study, only two chemical features were used: molecular weight and $XlogP$, which are easily accessible on PubChem. These features were combined with the similarity score developed by Tatonetti, to predict the permeability of the BBB to a particular agent, with features consisting of similarity to each of six expert-curated CNS-penetrant agents (antipsychotics and antidepressants). These similarity-based features led to a significant improvement in the performance of a regression model trained to predict BBB permeability (McCoy and Perlis, 2015). The authors report an area under the receiver operating characteristics curve (AUROC) of 0.65 with molecular weight and $XlogP$ features and 0.73 with addition of the statistically derived similarity score upon evaluation using a reference set created by Martins et al. (McCoy and Perlis, 2015; Martins, Teixeira, Pinheiro and Falcao, 2012).

While this work demonstrates the potential of side effect similarity derived from SDAs for drug repurposing, the most widely-used SDAs still have limitations in that they are surrogate measures of statistical association of drug-event pairs, and do not possess the capacity to model implicitly intrinsic effects of similar drugs (such as drug class effects) and side effects (for example, effects with similar mechanisms such as stroke and myocardial infarction). This is an innate limitation of SDAs, because these measures are derived from counts of report-level co-occurrence between discrete entities (Portanova, Murray, Mower, Subramanian and Cohen, 2020). In the NLP community, discrete representations of words have been largely superseded by distributed vector representations, which are trained on a large unlabeled corpus such that words that occur in similar contexts will have similar vector representations. This similarity is valuable for downstream machine learning applications, as it supports generalization, permitting models to recognize the same idea expressed in different words. The dichotomy between distributed and discrete word representations is analogous to that between discrete and distributed FAERS-derived drug representations, in that there is reason to believe generalization may be of value here also. If two drugs cause similar side effects, they may act on shared targets in a signaling pathway, as well as having similar therapeutic properties (Pushpakom et al., 2019; Gottlieb, Stein, Ruppinn and Sharan, 2011).

Recently, Portanova et al. introduced a novel representational approach called aer2vec, which derives distributed representations of data from FAERS that can be used to identify drug side-effect associations (Portanova et al., 2020). aer2vec was inspired by the skipgram-with-negative-sampling (SGNS) algorithm (Mikolov, Chen, Corrado and Dean, 2013a), which is implemented in the widely-used word2vec and fastText software packages. The SGNS algorithm trains a shallow neural network to predict $P(\text{context word} | \text{observed word})$, where context is defined by a sliding window centered on the observed word. This neural-probabilistic model can be adapted in the context of AERS to predict $P(ADE | \text{drug})$ (or $P(\text{drug} | ADE)$), with context defined as an ADE report. The results of this work demonstrated significant improvements in performance in identification of drug/side-effect relationships on two pharmacovigilance reference sets when using aer2vec representations, as compared with widely-used disproportionality metrics (Portanova et al., 2020), with performance of all models (including baseline models) improved by restricting training on the basis of provider designations of suspicion.

In this paper, we evaluate the utility of aer2vec representations of FAERS data for drug repurposing. Specifically,

we combine distributed aer2vec representations (drug embeddings) with chemical features to predict the permeability of the BBB to certain small molecules. In doing so, we adapt the approach of McCoy and Perlis (McCoy and Perlis, 2015), substituting distributed representations for the statistically-derived measures of similarity they employed (the measures from Tatonetti et al's previous work (Tatonetti, Fernald and Altman, 2012a)). We hypothesized that adding distributed drug vectors generated from FAERS data to the basic chemical properties would improve performance when identifying the permeability of the BBB to a drug, as compared with prior work where statistics concerning discrete occurrence counts in FAERS were used to estimate drug-drug similarity.

2. Methods

2.1. Training and evaluation data

2.1.1. AER Data

FAERS data were obtained from a normalized, standardized edition of the database developed by Banda et al. (Banda, Evans, Vanguri, Tatonetti, Ryan and Shah, 2016). The dataset spans from 2004 to 2015 and was standardized by mapping drug names to RxNorm and outcomes to SNOMED-CT (De Silva, MacDonald, Paterson, Sikdar and Cochrane, 2011; Liu, Ma, Moore, Ganesan and Nelson, 2005). As in the original aer2vec work (Portanova et al., 2020), three subsets of this dataset were generated as follows:

1. We retained the full dataset (FULL), with the inclusion of all the drugs and side effects in a report.
2. Using the role code variables primary suspect (PS) and secondary suspect (SS), which represent reporter designations of suspicion, we produced subsets that are composed of drugs in the PS role and their concomitant ADEs; and the disjunction of drugs in the PS and SS (PS-SS) roles and their ADEs.

Portanova et al. describe the development of two aer2vec neural embedding architectures (Portanova et al., 2020). The aer2vec+ model is trained to predict the probability of observing a drug given the observation of an ADE in a report ($P(\text{drug}|\text{ADE})$); and the aer2vec- model is trained to predict the probability of an ADE given the observation of a drug ($P(\text{ADE}|\text{drug})$). In this study, we evaluated aer2vec- rather than aer2vec+, because with neural word embeddings, neural network input weights are typically used to estimate semantic similarity, and drugs (rather than adverse events) have input weight vectors with aer2vec-.

2.1.2. Training aer2vec-

For each of the dataset configurations (FULL, PS, and PS-SS), we generated drug embeddings by training the aer2vec- model to predict $P(\text{ADE}|\text{drug})$ with the following training objective (Portanova et al., 2020) (equation 2):

$$\underset{\theta}{\operatorname{argmax}} \left[\sum_{(\text{drug}, \text{ADE}) \in R} \log \sigma(v_{\theta}(\text{drug}) \cdot v_{\theta}(\text{ADE})) + \sum_{(\text{drug}, \text{ADE}) \in R'} \log \sigma(-v_{\theta}(\text{drug}) \cdot v_{\theta}(\text{ADE})) \right] \quad (2)$$

Similarly to the original skip-gram architecture developed by Mikolov et al. (Mikolov, Sutskever, Chen, Corrado and Dean, 2013b), but substituting drugs and ADEs for observed and context words respectively, each drug is connected to a hidden layer by a vector of input weights (drug in Equation 2). The hidden layer is in turn connected to an output weight vector (ADE in Equation 2) for each ADE. Drugs and ADEs are observed in FAERS reports, R (the FULL set, or the PS/PS-SS subsets). ADE indicates events that were observed with a drug in a report, and $\neg \text{ADE}$ indicates randomly drawn counterexamples ("negative samples"), approximating the drug/event pairs that were not observed in any report (R'). σ represents the sigmoid function, that converts the scalar product of the input and output weight vectors for a particular drug/event pair to a value between zero and one that can be interpreted probabilistically. The model is trained to optimize this objective, using stochastic gradient descent. With aer2vec-, input weights for drugs are retained, permitting the estimation of similarity between a pair of drugs as the cosine of the angle between their respective input weight vector representations.

During training, we encoded all of the drug and ADE terms meeting the relevant constraints (PS/PS-SS/FULL), aside from a stoplist of 28 ADE terms (such as "NA", "test", "error") that were previously identified as lacking semantic content upon inspection of the data (Portanova et al., 2020). The vectors were trained over 25 iterations across the data with five negative samples per positive sample, and without subsampling of frequent terms (which is commonly applied when modeling language), as subsampling was shown to be detrimental to performance in pharmacovigilance

tasks (Portanova et al., 2020). In particular, the **aer2vec** model is trained to **maximize** the predicted **probabilities** for observed drug/ADE co-occurrence events (where a drug and ADE appear in the same **FAERS** report) while **minimizing** this predicted **probability** for randomly sampled **counterexamples** (the ‘negative samples’ in skipgram-with-negative-sampling). For each observed drug/ADE co-occurrence, we randomly draw five other ADEs, and minimize the predicted probability of their co-occurring with the drug in question. We used five negative samples per observed co-occurrence event, following Portanova et al (Portanova et al., 2020).

2.1.3. Reference Sets and Data Sources

The ground-truth assertion of **BBB permeability** was obtained from the supplementary data of Martin et al. (Martins et al., 2012), which includes ground-truth assertions for **2337 drugs**, with **1152** labeled as CNS-penetrant (**BBB+**) and **785** as non-penetrant (**BBB-**). The dataset was created by **compiling information** from **a range of different published studies** (Martins et al., 2012). A second reference dataset, provided by Gao et al. (Gao et al., 2017), was also used. This set contains **179 BBB+** and **145 BBB-** drugs (Gao et al., 2017) to test our models. After restricting the **reference sets** to those drugs that appear in both **FAERS** (a requirement for both the generation of FAERS-derived embeddings and calculation of the Tanimoto similarity scores following the method described in (Campillos et al., 2008; Tatonetti et al., 2012a)), as well as **OffSides**, we ensured **mutual exclusivity** between the **two reference sets** by removing any overlapping drugs from the larger of the two sets only. The **OffSides** database has **1325 unique drugs**, and had **414** drugs and **215** drugs **overlapping with Martin’s** and **Gao’s sets** respectively. Another restraining factor is that the drugs **must** have **embedding** representations, which are **derived from FAERS**. As a result, we **retained 188** drugs from **Martin’s set** (125/63 **BBB+/BBB-**) and **208** (128/80 **BBB+/BBB-**) drugs from **Gao’s set** (both datasets have more positive samples than negative samples). **The majority (~90%)** of drugs in the **Martin et al.** set were **not eligible** for inclusion for two reasons. **Firstly**, the drugs have to appear in both **OffSides FAERS**. Secondly, any drugs appearing in the reference set **from Gao et al.** were subsequently **excluded**.

Following McCoy and Perlis (McCoy and Perlis, 2015), the chemical properties of lipophilicity ($XlogP$) and molecular weight (MW) of these drugs were acquired from PubChem (Bolton, Wang, Thiessen and Bryant, 2008), using the Python package **pubchempy 2017** (Swain, Sjogren, Bjorn, Lazzaro and Hsiao, 2017). These properties were selected because previous research has demonstrated the permeability to chemical compounds is associated with both their lipophilicity and their MW (McCoy and Perlis, 2015; Wang et al., 2018). Since the mean values of MW and $XlogP$ had different scales, we applied a Z transformation across all features, except for the **distributed** vector representations.

2.1.4. Drug Panel

As a point of comparison, we reproduced the method of feature engineering described in (McCoy and Perlis, 2015). This involved calculating similarities to members of a **six-drug** panel of the expert-curated CNS-penetrant agents, comprising of antipsychotics: haloperidol, olanzapine and risperidone; and antidepressants: fluoxetine, mirtazapine and nortriptyline. Following this work, we estimated similarities between agents in our reference sets and each drug in this six-drug panel. We used the **OffSides** dataset developed by Tatonetti and his colleagues¹ and calculated the **Tanimoto similarity scores** between the drug in the two reference lists and the **six-panel** drugs following equation (1), as exemplars of statistically-derived similarity scores (Tatonetti et al., 2012a; Campillos et al., 2008). This resulted in a set of **six additional features** for **each drug** in our reference sets.

In addition to these **statistically-derived** similarities, we estimated the **cosine similarity** between drug panel and **reference set drugs** using our **aer2vec- drug embeddings**, resulting in a **further six features** for each drug which provide the basis for a direct comparison between the utility of **drug-drug similarities** derived from **discrete** statistics and those from **aer2vec- distributed representations**. In some cases, we included the **500-dimensional distributed** representations as **features** directly.

2.2. Classification and Model Configuration

To compare the utility of **discrete** and **distributed** representations as a means to predict BBB penetration, we adapted the approach of **McCoy and Perlis** (McCoy and Perlis, 2015), which involved training **LR models** on **feature sets** that included molecular features ($XlogP$ and MW) with and without the addition of features derived from similarity to drugs in the **six drug panel**. We evaluated the utility of including the **drug embedding** itself **as a feature vector**, with the hypothesis that this may provide useful information beyond that captured by the drug-to-panel similarities.

¹available from <http://tatonettilab.org/resources/nsides/>

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LR: Logistic Regression

random forest

feature selection

Predicting Drug Penetrance to Blood Brain Barrier

Our experiments occurred in three phases. In the first phase, we trained drug embeddings using the aer2vec-model with a dimension of 500 following (Portanova et al., 2020), and using the FULL dataset (i.e. considering all drugs in a report irrespective of provider designation of suspicion) derived from a normalized standardized edition of the FAERS data (Banda et al., 2016). We picked the FULL dataset to begin with as it is the most comprehensive. Drugs in the reference sets were pre-labeled with the ground-truth assertion of BBB permeability. A LR model with L2 regularization was fitted on the following three feature sets, including combinations with Z transformation:

1. chemical features (CF) from PubChem: lipophilicity (XlogP) and molecular weight (MW)
2. similarity-based features (using the six-drug panel) based on statistical associations using the Tanimoto similarities (SIM_{Tani}) (Tatonetti et al., 2012a)
3. similarity-based features (using the six-drug panel) from comparing aer2vec-drug embeddings (SIM_{a2v})
4. the aer2vec-drug embeddings themselves (VEC_{a2v})

We trained a LR model on each of the features separately, and all the features collectively. Models were evaluated using stratified 5-fold cross-validation (5CV). In order to evaluate the robustness of the model, we also implemented a cross-training configuration by training on the larger reference set (Gao et al., 2017) and testing on the smaller one (Martins et al., 2012) (the latter set became the smaller of the two once overlapping drugs were removed from it). The rationale for choosing LR (aside from its use in (McCoy and Perlis, 2015)) was that it is parametric and scales readily to large datasets. In addition, LR is found in a wide variety of deep learning applications as a final layer, and our prior work indicates that LR models can be effectively applied to literature-derived distributed representations for accurate prediction of drug effects (Mower, Subramanian, Shang and Cohen, 2016). Which is to say, LR models can support accurate classification in the 5CV was chosen for consistency with other prior research with similar topics. To assess performance, the AUROC was computed within (5CV) and across (cross-training) reference sets.

Secondly, we experimented with embeddings of lower dimensionality. Our objective was to rule out the possibility that the best performance observed in our initial experiments was a consequence of the larger number of features - and hence trainable parameters in the VEC_{a2v} regression models (502 vs. 8). To further investigate this possibility, and to ameliorate the problem of overfitting, principle component analysis (PCA) was applied to reduce the dimensionality of the original 500-dimensional vectors. The first six principal components were retained in the analyses. To eliminate the potential influence of differences in dimensionality, we used the 6-drug panel described above as a point of comparison for the performance of the 6-component PCA. The corresponding feature sets were as follows:

1. the 6 principal components derived from a PCA of all the available drug embeddings resulting in six features for each drug
2. the matrix of 6 individual embedding-derived panel scores.

L2-regularized models was utilized because L2-regularization can minimize prediction errors with dense vector representations where most components are non-zero. PCA eliminates the effect of those features that have less contribution to the variance in the datasets.

Finally, to evaluate the utility of rater designations of suspicion for this task, we constructed models using aer2vec-vectors derived from the FULL, PS, and PS-SS data sets. In doing so, our goal was to understand given the aer2vec-model, which dataset gives us the best predictive modeling performance, motivated by the utility these provider designations of suspicion had shown in improving predictive modeling of drug/ADE relationships (Figure 1).

To assess the reliability of the results, all the experiments were repeated 20 times and a 95% confidence interval was calculated along with the results of 5CV. The models were created using sci-kit learn (version 0.19.075) and the Anaconda distribution of Python (version 3.6.1). To generate aer2vec embeddings we used the open-source implementation of this method that is publicly available as a component of the Semantic Vectors software package for distributional semantics research (Widdows and Cohen, 2010; Portanova et al., 2020)².

3. Experiments and Results

Using L2-regularized LR on the two datasets, our 5CV results show that with the addition of six individual, statistically-derived similarity scores, the AUC improves from a baseline with chemical features only. Upon replacing the six individual Tanimoto similarity scores with embedding-derived panel scores, the AUC improves even further.

²available from <https://github.com/semanticvectors/semanticvectors.git>, with the procedure for creating aer2vec drug embeddings described at <https://github.com/treversec/aer2vec.git>

Predicting Drug Penetrance to Blood Brain Barrier

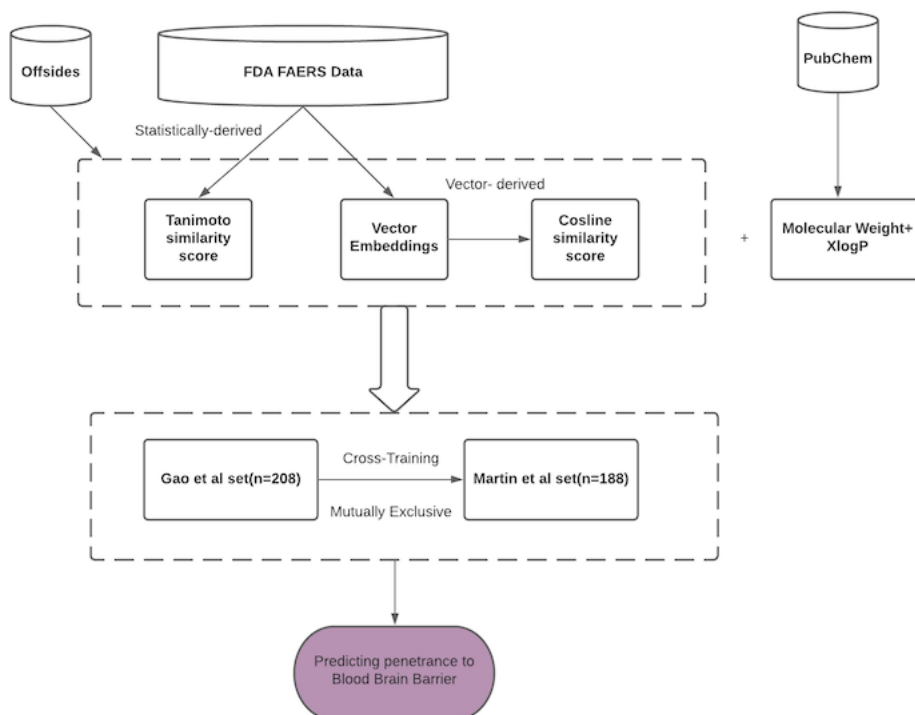


Figure 1: Flowchart of the experiments.

Table 1. 5CV and cross-training results. Vector-derived similarity score is the cosine similarity (normalized dot product) between the panel score and vectors. Mean AUC from 5CV and 95% Confidence Interval using 500-dimensional aer2vec embeddings trained on the FAERS data. (Martin: n=188; Gao: n=208). SIM_{Tani} : Tanimoto similarity score, SIM_{a2v} : Cosine similarity score, VEC_{a2v} : aer2vec vectors.

Models	5CV Martins et al. dataset	5CV Gao et al. dataset	Train:Gao; Test:Martin
MW + xlogP	0.719 (0.719, 0.720)	0.741 (0.740, 0.742)	0.728
MW + xlogP + SIM_{Tani}	0.758 (0.756, 0.760)	0.780 (0.780, 0.780)	0.755
MW + xlogP + SIM_{a2v}	0.763 (0.762, 0.764)	0.821 (0.821, 0.821)	0.783
MW + xlogP + 500D VEC_{a2v}	0.812 (0.811, 0.813)	0.873 (0.871, 0.875)	0.788
500D VEC_{a2v}	0.801 (0.800, 0.803)	0.825 (0.823, 0.826)	0.764

Lastly, we observe an additional increase of the AUC to 0.812 in Martins et al's dataset and to 0.873 in Gao et al's dataset when replacing the similarity scores with the FULL 500-dimensional aer2vec vectors. Using 500-dimensional semantic vectors alone, the performance is only slightly worse than with the combination of chemical features with these vectors, and considerably better than with the chemical features alone. In the cross-training configuration, the same trends were observed as with the two reference sets. The baseline AUC is increased from 0.728 to 0.788 when using vector representations in addition to chemical features (Figure 1, Table 1).

With six-dimensional features, we found that all models including embeddings (or their reduced-dimensional approximations) showed superior performance to the models based on similarity to the six-drug panel. These improvements in performance were retained when limiting the size of the embedding models to 6 dimensions, demonstrating superior performance at the same number of trainable parameters. For both sets and in the cross-training configuration, the best performance amongst these reduced-dimensional models was observed in the model with chemical features augmented by six PCA features derived from 500-dimensional vectors (Table 2).

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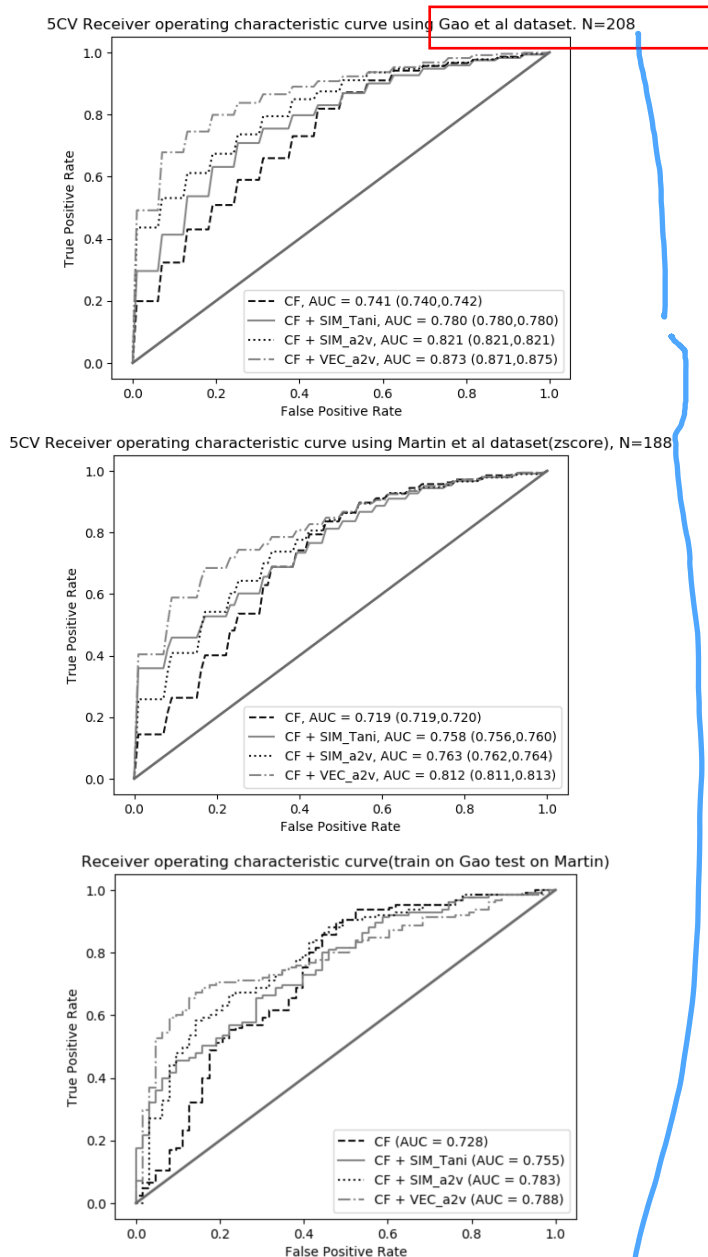


Figure 2: Results with FULL FAERS set with 500-dim vectors in 5 fold CV. **Top:** Martins et al. dataset; **Middle:** Gao et al. dataset. **Bottom:** cross-training configuration, trained on Gao and tested on Martin. ROC curves for CNS penetration including chemical features alone, with discrete statistically-derived Tanimoto scores and with embedding-derived similarity panel scores. The mean and the confidence intervals were calculated across 20 repeated cross-validation splits. CF: chemical features, SIM_{Tani} : Tanimoto similarity score, SIM_{a2v} : Cosine similarity score, VEC_{a2v} : aer2vec vectors.

When comparing the three FAERS-derived datasets (PS, PS-SS and FULL) across all models, we observed that models trained on the PS-SS data had the best performance when embeddings were used as features, with the single exception of Gao et al's dataset with 500-dimensional vectors (Table 3), where vectors derived from the FULL set performed best. In the cross-training configuration, models using PS-SS data performed best.

Table 2. Mean 5CV AUC with 6-dimensional vectors derived from FULL within two reference sets and the cross-training configuration. 6 principle components were reduced from 500 dimensional semantic representations.

Model	5CV Martin et. al. dataset	5CV Gao et. al. dataset	Train:Gao;Test:Martin
MW + xlogP + SIM_{Tani}	0.758 (0.756, 0.760)	0.780 (0.780, 0.781)	0.755
MW + xlogP + 6-dimensional SIM_{a2v}	0.774 (0.771, 0.778)	0.850 (0.847, 0.852)	0.792
MW + xlogP + 6 principal components	0.798 (0.793, 0.803)	0.872 (0.871, 0.873)	0.827

Table 3. Comparison of 5CV(L2-regularized) of vector representation derived from PS, PS-SS and FULL with 500 dimensions, when added to chemical features within two reference sets as well as in the cross-training configuration.

Model	Martin et al. dataset	Gao et al. dataset	Train: Gao; Test: Martin
PS 500D	0.812 (0.805, 0.820)	0.825 (0.820, 0.831)	0.797
PS-SS 500D	0.846 (0.842, 0.850)	0.856 (0.852, 0.861)	0.839
FULL 500D	0.812 (0.811, 0.813)	0.873 (0.871, 0.875)	0.788

4. Discussion

In this study, we used chemical features in addition to the vector representations derived from the aer2vec-embeddings as features for a LR classifier trained to predict the permeability of the BBB to certain drugs. We found that substituting similarities between these embeddings for discrete statistically-derived drug-to-panel similarities used in previous work improved performance, and that including the embeddings themselves as features (with or without additional dimension reduction to normalize the number of trainable parameters across models) improved performance even further. These results suggest that the aer2vec- model captures information from AERs that has value for drug repurposing, in addition to the previously demonstrated utility of this model for pharmacovigilance signal detection (Portanova et al., 2020). Our results are aligned with findings from previous work, in that adding the discrete statistically-derived similarity score improved the performance of the LR classifier (McCoy and Perlis, 2015) in both cross-validation and cross-training configurations. However, a notable difference in methodology is that we used the individual panel similarity scores as six features, rather than the mean score across the six-drug panel as a single feature. Unlike the original study, we found that taking the average across the six statistically-derived panel similarities did not reliably improve model performance. One explanation for this inconsistency might be that the drugs we used in our study differ from those used in McCoy and Perlis's work. As the reference set of drugs used in their work is not publicly available, we are unable to further investigate this disparity. However, another possible explanation may concern our use of a newer version of the OffSides database, published in 2019, to calculate the Tanimoto coefficients. McCoy and Perlis used an earlier version which is no longer publicly available.

Notably, similarities derived by comparing distributed representations of reference set drugs to those of drugs in the 6-drug panel appear to be even more effective as features than the discrete statistical estimates from Tatonetti et al (Tatonetti et al., 2012a). We explored the difference between the distributions of the two sets of similarity scores in an effort to explain this observation. Across all reference set drugs, the statistically derived similarity scores are normally distributed and the distributions of the BBB+ and BBB- drugs are homogeneous. In contrast with aer2vec-derived cosine similarities the distribution skewed further right, and the distributions of the BBB+ and BBB- drugs are more heterogeneous (Supplement Figures 1 and 2). This difference in the distribution of similarity scores across positive and negative classes may explain why the cosine similarities provide a better basis for classification.

Both this difference in distribution and the substantial improvement in AUC may be attributable to the difference in the methods of calculating the similarity scores. The similarities used by McCoy and Perlis were estimated using the Tanimoto coefficient (McCoy and Perlis, 2015; Tatonetti et al., 2012a). We calculated our similarity scores using the cosine distance between the embeddings representing the six-drug panel and drugs in our reference sets. Literature suggests that cosine similarity in and of itself can compare favorably to other similarity measure techniques, including the Tanimoto coefficient (Ariff, Bakar and Rahmad, 2018). In our case, cosine similarity is derived from aer2vec embeddings, which have the additional property of generalization, such that drugs with similar but not identical side effects may develop similar representations. This distinction between discrete and distributed representations may be largely responsible for the differences in performance observed in these studies, as well as in previous work on pharmacovigilance signal detection (Portanova et al., 2020). However, further research is required to better characterize the ways in which this generalization contributes to enhanced performance in both cases.

Simply replacing the Tanimoto coefficients with cosine similarities between the aer2vec-embeddings resulted in improved performance (Table 1). However, replacing the six panel-derived cosine similarities with the embeddings themselves led to even better performance in both datasets. One might argue that this additional boost in performance can be attributed to the addition of 500 trainable parameters, and as such is simply an artifact, and that these additional parameters also come at higher computational cost and increase model complexity and the danger of overfitting (Wu, Schuster, Chen, Le, Norouzi, Macherey, Krikun, Cao, Gao, Macherey, Klingner, Shah, Johnson, Liu, Kaiser, Gouws, Kato, Kudo, Kazawa, Stevens, Kurian, Patil, Wang, Young, Smith, Riesa, Rudnick, Vinyals, Corrado, Hughes and Dean, 2016). However, this is not the case in our study, as even when the vectors are reduced to six dimensions using PCA, the improvements in performance are sustained (Table 1 and Table 2). In some cases, models with six PCA components derived from 500-dimensional vectors as features performed better than those using the original vectors themselves. This was the evident in cross-validation on one of two datasets (Gao et al.'s), and more substantially in the cross-training configuration (Table 2). Using derived principal components also appears to offer an advantage over simply training 6-dimensional aer2vec embeddings, which is what one might anticipate on account of PCA's capability to preserve the information that is of greatest value in distinguishing between points in a higher dimensional space. Furthermore, PCA finds the components that best preserve contrast within a subset of data, where a 6-dimensional hidden layer is trying to maintain contrast across all data points. Additionally, reducing dimensionality with PCA can eliminate noise - which in this case could be introduced by encoding of contextual features that are not relevant to this classification task.

Furthermore, when comparing embeddings derived from the PS (drugs designated as primary suspects), PS-SS (drugs designated as primary or secondary suspects), and FULL (all drugs in a report) data sets, we originally hypothesized that embeddings derived from PS would lead to best performance, as they did in pharmacovigilance signal detection tasks Portanova et al. (2020). However, we consistently observed that embeddings trained on the PS-SS set led to best performance using Martin's reference set as well as in the cross-training configuration. In contrast, with the dataset from Gao et al. performance continued to improve as more data were included, with best performance with the full set. The PS-SS subset retains more information than the PS subset for both sets.

Regarding dimensionality, we did not observe a consistent pattern across the two reference sets. With the dataset from Gao et al., performance is retained at lower dimensionality. While it seems intuitive that higher dimensional embeddings would accommodate more information, it has been suggested that excessive dimensionality can introduce additional noise into a model, resulting in inconsistent performance and a danger of overfitting (Table 2, 3) (Yin and Shen, 2018; Mower et al., 2016). This proposal is consistent with the improvement in performance on the Martin set observed at lower dimensionality in the cross-training configuration only, with a loss in performance at low dimensionality on this set in a cross-validation scenario. In this study, the pattern is inconclusive and further investigations may be needed to determine the optimal dimensionality for this task.

Overall, our results demonstrate that the models including FAERS based vector embeddings outperformed both the models with chemical features only and those with chemical features and similarity scores (Gao et al., 2017; McCoy and Perlis, 2015). Even when using semantic vectors alone without chemical features, the performance of predicting the penetrance of the BBB is only slightly lower than that of the model with semantic representation+chemical features (Table 1). While our results are not strictly comparable to those from previously published studies on account of differences in data sets and drug coverage, we include a discussion of prior results to provide some additional context. Two recent publications using machine learning methods to predict the permeability of small molecules to BBB report AUROCs of 0.919 and 0.98 respectively (Wang et al., 2018; Miao et al., 2019). In Wang et al's studies, various machine learning methods were applied to a larger set of chemical features including MW, xlogP, topological polar surface area (TPSA), and other 2D molecular descriptors (including structures) and molecular fingerprints of 2213 molecules. Aside from this expanded set of features, a possible explanation for their reported performance is that the distribution of BBB+ and BBB- in their reference sets was highly imbalanced. Another may involve the algorithms used for classification of these features. LR was one of the models used, however, the results with LR were not reported, because the other models reportedly outperformed it (Wang et al., 2018). Moreover, in Miao et al's study, the methods proposed in Wang et al. were adapted with additional development of a four-layer deep neural network (DNN), resulting in a reported AUC of 0.98 (Miao et al., 2019). Unfortunately, neither the data sets nor the code used in these experiments was not provided. Consequently we are unable to reproduce these results, nor are we able to evaluate their method against others on a common reference set. Nonetheless, the results as reported do suggest that deeper neural network models may improve performance over LR on this task. As deep neural network models could be readily applied to our aer2vec-derived features, this is a direction we plan to explore in future work.

One important limitation of our work is that the distributed representations were derived from a standardized edition of the FAERS data and we have not evaluated our methods using “raw” FAERS data as a basis for training embeddings. It is an open question as to what extent the generalization capabilities of aer2vec might compensate for the lack of explicit terminologically-driven standardization of drug names in FAERS. A related limitation is that the standardized FAERS data set concludes with reports from the year 2015. Including additional data beyond this point may lead to improvements in performance. Finally, even though our models achieved high accuracy in predicting BBB permeability, *in vitro* models would still be needed to validate predictions beyond the available reference sets to establish the practical utility of these models for drug repurposing.

In future research, we plan to address these limitations and extend our methods in three ways. Firstly, we intend to apply additional chemical features, such as hydrogen bond donors and receptors, and other features used in prior work (Gao et al., 2017; Doniger et al., 2002; Wang et al., 2018; Miao et al., 2019) to the model. Secondly, we plan to evaluate more sophisticated machine learning classification methods to evaluate their utility when trained on aer2vec embeddings and other features. Finally, we will evaluate the utility of other drug embeddings for this task, including those trained on the biomedical literature.

5. Conclusion

In this study, we evaluated the utility of aer2vec-distributed representations of AER data as an adjunct to chemical features for predicting drug molecule penetration of the BBB. Within and across two data sets, our approach outperformed previously established methods when embedding-derived similarity was substituted for statistically-derived estimates of similarity from FAERS, and even more so when the embeddings themselves were used directly as features. We further explored the role of dimensionality in the evaluation and observed that the performance of the model was enhanced relative to prior work even when using low-dimensional embeddings as features, leading us to the conclusion that the improved performance cannot be attributed to differences in the number of trainable parameters across LR models. These findings support the proposal that distributed representations of AER data have utility for predicting permeability of the BBB in particular, and for drug repurposing in general.

References

- Ariff, N.M., Bakar, M.A.A., Rahmad, M.I., 2018. Comparative study of document clustering algorithms. *International Journal of Engineering Technology* 7, 246. doi:10.14419/ijet.v7i4.11.20816.
- Ashburn, T.T., Thor, K.B., 2004. Drug repositioning: identifying and developing new uses for existing drugs. *Nat Rev Drug Discov* 3, 673–683.
- Banda, J.M., Evans, L., Vanguri, R.S., Tatonetti, N.P., Ryan, P.B., Shah, N.H., 2016. A curated and standardized adverse drug event resource to accelerate drug safety research. *Sci Data* 3, 160026.
- Bate, A., Evans, S.J., 2009. Quantitative signal detection using spontaneous ADR reporting. *Pharmacoepidemiol Drug Saf* 18, 427–436.
- Bergstrom, C.A., Charman, S.A., Nicolazzo, J.A., 2012. Computational prediction of CNS drug exposure based on a novel *in vivo* dataset. *Pharm. Res.* 29, 3131–3142.
- de Bie, S., Ferrajolo, C., Straus, S.M., Verhamme, K.M., Bonhoeffer, J., Wong, I.C., Sturkenboom, M.C., Sturkenboom, M., Bonhoeffer, J., Gi-aquinto, C., Gazarian, M., Ceci, A., Weereasuriya, K., Hirschfeld, S., Saint Raymond, A., Nakamura, H., Sharland, M., Offringa, M., Wong, I.C., Hoppu, K., 2015. Pediatric Drug Safety Surveillance in FDA-AERS: A Description of Adverse Events from GRiP Project. *PLoS ONE* 10, e0130399.
- Bolton, E.E., Wang, Y., Thiessen, P.A., Bryant, S.H., 2008. Pubchem: integrated platform of small molecules and biological activities, in: *Annual reports in computational chemistry*. Elsevier. volume 4, pp. 217–241.
- Campillos, M., Kuhn, M., Gavin, A.C., Jensen, L.J., Bork, P., 2008. Drug target identification using side-effect similarity. *Science* 321, 263–266.
- De Silva, T.S., MacDonald, D., Paterson, G., Sikdar, K.C., Cochrane, B., 2011. Systematized nomenclature of medicine clinical terms (snomed ct) to represent computed tomography procedures. *Comput. Methods Prog. Biomed.* 101, 324–329. URL: <https://doi.org/10.1016/j.cmpb.2011.01.002>, doi:10.1016/j.cmpb.2011.01.002.
- Doniger, G.M., Foxe, J.J., Murray, M.M., Higgins, B.A., Javitt, D.C., 2002. Impaired visual object recognition and dorsal/ventral stream interaction in schizophrenia. *Arch. Gen. Psychiatry* 59, 1011–1020.
- Edwards, I.R., Aronson, J.K., 2000. Adverse drug reactions: definitions, diagnosis, and management. *Lancet* 356, 1255–1259.
- Gao, Z., Chen, Y., Cai, X., Xu, R., 2017. Predict drug permeability to blood-brain-barrier from clinical phenotypes: drug side effects and drug indications. *Bioinformatics* 33, 901–908.
- Gottlieb, A., Stein, G.Y., Rupp, E., Sharan, R., 2011. PREDICT: a method for inferring novel drug indications with application to personalized medicine. *Mol. Syst. Biol.* 7, 496.
- Harpaz, R., DuMouchel, W., LePend, P., Bauer-Mehren, A., Ryan, P., Shah, N.H., 2013. Performance of pharmacovigilance signal-detection algorithms for the FDA adverse event reporting system. *Clin. Pharmacol. Ther.* 93, 539–546.
- Heckman-Stoddard, B.M., DeCensi, A., Sahasrabudhe, V.V., Ford, L.G., 2017. Repurposing metformin for the prevention of cancer and cancer recurrence. *Diabetologia* 60, 1639–1647.

- Hennessy, S., Strom, B.L., 2015. Improving postapproval drug safety surveillance: getting better information sooner. *Annu. Rev. Pharmacol. Toxicol.* 55, 75–87.
- Kortagere, S., Chekmarev, D., Welsh, W.J., Ekins, S., 2008. New predictive models for blood-brain barrier permeability of drug-like molecules. *Pharm. Res.* 25, 1836–1845.
- Kumar, R., Harilal, S., Gupta, S.V., Jose, J., Thomas Parambi, D.G., Uddin, M.S., Shah, M.A., Mathew, B., 2019. Exploring the new horizons of drug repurposing: A vital tool for turning hard work into smart work. *Eur J Med Chem* 182, 111602.
- Liu, S., Ma, W., Moore, R., Ganesan, V., Nelson, S., 2005. Rxnorm: Prescription for electronic drug information exchange. *IT Professional* 7, 17–23. URL: <https://doi.org/10.1109/MITP.2005.122>, doi:10.1109/MITP.2005.122.
- Martins, I.F., Teixeira, A.L., Pinheiro, L., Falcao, A.O., 2012. A Bayesian approach to in silico blood-brain barrier penetration modeling. *J Chem Inf Model* 52, 1686–1697.
- McCoy, T.H., Perlis, R.H., 2015. A tool to utilize adverse effect profiles to identify brain-active medications for repurposing. *Int. J. Neuropsychopharmacol.* 18.
- Miao, R., Xia, L.Y., Chen, H.H., Huang, H.H., Liang, Y., 2019. Improved Classification of Blood-Brain-Barrier Drugs Using Deep Learning. *Sci Rep* 9, 8802.
- Mikolov, T., Chen, K., Corrado, G., Dean, J., 2013a. Efficient estimation of word representations in vector space. *arXiv:1301.3781*.
- Mikolov, T., Sutskever, I., Chen, K., Corrado, G., Dean, J., 2013b. Distributed representations of words and phrases and their compositionality, in: *Proceedings of the 26th International Conference on Neural Information Processing Systems* - Volume 2, Curran Associates Inc., Red Hook, NY, USA. p. 3111–3119.
- Moore, T.J., Zhang, H., Anderson, G., Alexander, G.C., 2018. Estimated Costs of Pivotal Trials for Novel Therapeutic Agents Approved by the US Food and Drug Administration, 2015–2016. *JAMA Intern Med* 178, 1451–1457.
- Mower, J., Subramanian, D., Shang, N., Cohen, T., 2016. Classification-by-Analogy: Using Vector Representations of Implicit Relationships to Identify Plausibly Causal Drug/Side-effect Relationships. *AMIA Annu Symp Proc* 2016, 1940–1949.
- Portanova, J., Murray, N., Mower, J., Subramanian, D., Cohen, T., 2020. aer2vec: Distributed Representations of Adverse Event Reporting System Data as a Means to Identify Drug/Side-Effect Associations. *AMIA Annual Symposium Proceedings* , 717–726.
- Pushpakom, S., Iorio, F., Eyers, P.A., Escott, K.J., Hopper, S., Wells, A., Doig, A., Williams, T., Latimer, J., McNamee, C., Norris, A., Sanseau, P., Cavalla, D., Pirmohamed, M., 2019. Drug repurposing: progress, challenges and recommendations. *Nat Rev Drug Discov* 18, 41–58.
- Rena, G., Lang, C.C., 2018. Repurposing Metformin for Cardiovascular Disease. *Circulation* 137, 422–424.
- Rogers, D.J., Tanimoto, T.T., 1960. A Computer Program for Classifying Plants. *Science* 132, 1115–1118.
- Swain, M., Sjogren, R., Bjorn, Lazzaro, L., Hsiao, Y., 2017. Pubchempy-python wrapper for the pubchem pug rest api. URL: <https://pubchempy.readthedocs.io/en/latest/guide/introduction.html#how-pubchempy-works>.
- Tatonetti, N.P., Fernald, G.H., Altman, R.B., 2012a. A novel signal detection algorithm for identifying hidden drug-drug interactions in adverse event reports. *J Am Med Inform Assoc* 19, 79–85.
- Tatonetti, N.P., Ye, P.P., Daneshjou, R., Altman, R.B., 2012b. Data-driven prediction of drug effects and interactions. *Sci Transl Med* 4, 125ra31.
- Wang, K., Wan, M., Wang, R.S., Weng, Z., 2016. Opportunities for Web-based Drug Repositioning: Searching for Potential Antihypertensive Agents with Hypotension Adverse Events. *J Med Internet Res* 18, e76.
- Wang, Z., Yang, H., Wu, Z., Wang, T., Li, W., Tang, Y., Liu, G., 2018. In Silico Prediction of Blood-Brain Barrier Permeability of Compounds by Machine Learning and Resampling Methods. *ChemMedChem* 13, 2189–2201.
- Widdows, D., Cohen, T., 2010. The semantic vectors package: New algorithms and public tools for distributional semantics, in: *2010 IEEE Fourth International Conference on Semantic Computing*, pp. 9–15.
- Wu, Y., Schuster, M., Chen, Z., Le, Q.V., Norouzi, M., Macherey, W., Krikun, M., Cao, Y., Gao, Q., Macherey, K., Klingner, J., Shah, A., Johnson, M., Liu, X., Kaiser, , Gouws, S., Kato, Y., Kudo, T., Kazawa, H., Stevens, K., Kurian, G., Patil, N., Wang, W., Young, C., Smith, J., Riesa, J., Rudnick, A., Vinyals, O., Corrado, G., Hughes, M., Dean, J., 2016. Google’s neural machine translation system: Bridging the gap between human and machine translation. *arXiv:1609.08144*.
- Yin, Z., Shen, Y., 2018. On the dimensionality of word embedding, in: *Bengio, S., Wallach, H., Larochelle, H., Grauman, K., Cesa-Bianchi, N., Garnett, R. (Eds.), Advances in Neural Information Processing Systems 31*. Curran Associates, Inc., pp. 895–906. URL: <http://papers.nips.cc/paper/7368-on-the-dimensionality-of-word-embedding.pdf>.

CRedit authorship contribution statement

YiFan Wu: Conceptualization of this study, Methodology, Writing - Original draft preparation. **Justin Mower:** Reviewed and edited the draft. **Devika Subramanian:** Reviewed and edited the draft.. **Trevor Cohen:** Conceptualization of this study, Methodology, reviewed and edited the draft..

A. Appendix

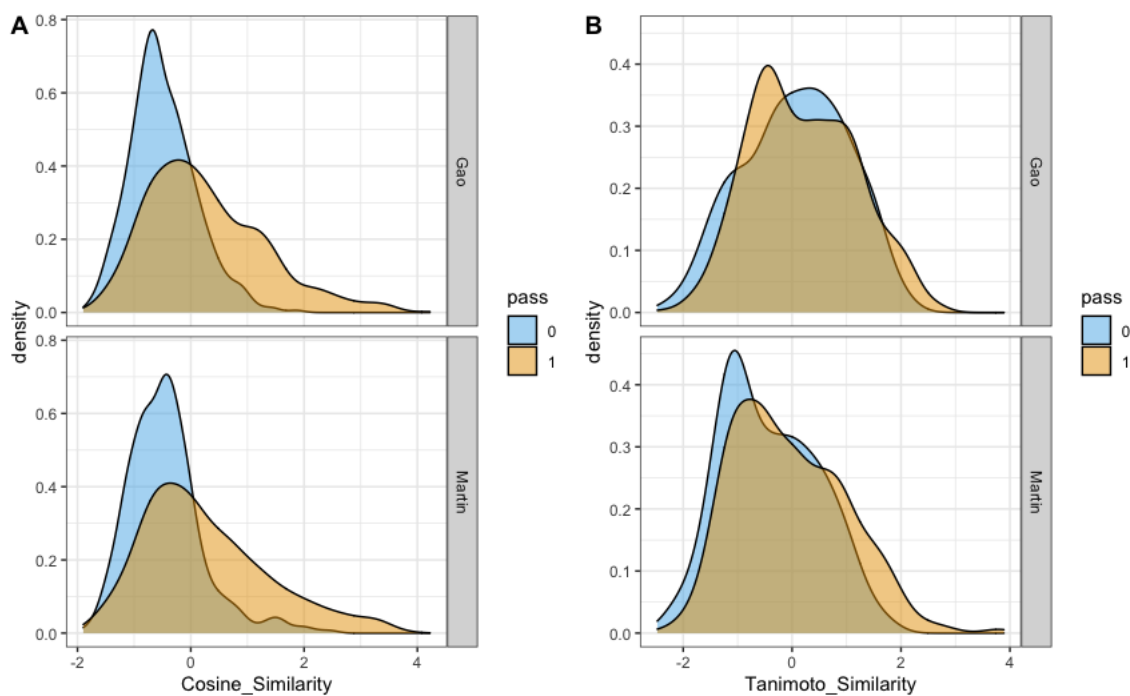


Figure 3: The density plot of Z-transformed cosine similarity and Tanimoto similarity grouping by BBB permeability in the two reference sets.